Reducibility 02

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Turing Machine

- Consider the following Turing machine:

  $M_1 =$ "On input x:

  1. *accept*"
- What is the language of TM M_1?
  - $L(M_1) = \Sigma^*$
- Why?

Action	Result
Run M_1 on input ε	*accept*
Run M_1 on input 0	*accept*
Run M_1 on input 1	*accept*
Run M_1 on input 00	*accept*
Run M_1 on input 01	*accept*
$\vdots$	$\vdots$

  - Because it accepts all strings

Turing Machine

- Consider the following Turing machine:

  $M_2 =$ "On input x:

  1. *reject*"
- What is the language of TM M_2?
  - $L(M_2) = \emptyset$
- Why?

Action	Result
Run M_2 on input ε	*reject*
Run M_2 on input 0	*reject*
Run M_2 on input 1	*reject*
Run M_2 on input 00	*reject*
Run M_2 on input 01	*reject*
$\vdots$	$\vdots$

  - Because it rejects all strings

Turing Machine

- Consider the following Turing machine:

  $M_3 =$ "On input x:

  1. Enter infinite loop"
- What is the language of TM M_3?
  - $L(M_3) = \emptyset$
- Why?

Action	Result
Run M_3 on input ε	Loops indefinitely
Run M_3 on input 0	Loops indefinitely
Run M_3 on input 1	Loops indefinitely
Run M_3 on input 00	Loops indefinitely
Run M_3 on input 01	Loops indefinitely
$\vdots$	$\vdots$

  - Because it loops indefinitely on all strings
- It does not accept or reject on all strings
- The language of a TM is the set of all strings **accepted** by it

Turing Machine

- Consider the following Turing machine:

  $M_4 =$ "On input x:
  1. If x is in the form of 0^n1^n, *accept*.
  2. If x is not in the form of 0^n1^n, *reject*."
- What is the language of TM M_4?
  - $L(M_4) = \{0^n1^n \mid n \geq 0\}$
- Why?

Action	Result
Run M_4 on input ε	*accept*
Run M_4 on input 0	*reject*
Run M_4 on input 1	*reject*
Run M_4 on input 00	*reject*
Run M_4 on input 01	*accept*
Run M_4 on input 10	*reject*
$\vdots$	$\vdots$

  - TM M_4 only accepts ε, 01, 0011, 000111, 00001111, ...

Turing Machine

- Consider the following Turing machine:

  $M_5 =$ "On input x:
  1. If x starts with a 0, $accept$.
  2. Enter infinite loop."
- What is the language of TM M_5?
  - $L(M_5) = 0\Sigma^*$
- Why?

Action	Result
Run M_5 on input ε	Loops indefinitely
Run M_5 on input 0	$accept$
Run M_5 on input 1	Loops indefinitely
Run M_5 on input 00	$accept$
Run M_5 on input 01	$accept$
Run M_5 on input 10	Loops indefinitely
$\vdots$	$\vdots$

  - TM M_5 only accepts 0, 00, 01, 000, 001, 010, 011, 0000, $\ldots$

Turing Machine

- Consider the following Turing machine:

  $M_6 =$ "On input x:
  1. If $x = \varepsilon$, $accept$.
  2. $reject$."
- What is the language of TM M_6?
  - $L(M_6) = \{\varepsilon\}$
- Why?

Action	Result
Run M_6 on input ε	$accept$
Run M_6 on input 0	$reject$
Run M_6 on input 1	$reject$
Run M_6 on input 00	$reject$
Run M_6 on input 01	$reject$
Run M_6 on input 10	$reject$
$\vdots$	$\vdots$

  - TM M_6 only accepts ε

Turing Machine

- Consider the following Turing machine:

  $M_7 =$ "On input x:
  1. If $x \neq \varepsilon$, $accept$.
  2. $reject$."
- What is the language of TM M_7?
  - $L(M_7) = \Sigma^* - \{\varepsilon\}$
- Why?

Action	Result
Run M_7 on input ε	$reject$
Run M_7 on input 0	$accept$
Run M_7 on input 1	$accept$
Run M_7 on input 00	$accept$
Run M_7 on input 01	$accept$
Run M_7 on input 10	$accept$
$\vdots$	$\vdots$

  - TM M_7 accepts any string that is not the empty string

Turing Machine

- Given a TM M and a string w, answer the following questions:
  1. What is $L(M)$?
     - We have no idea
  2. Is M a decider?
     - We have no idea
  3. What is the result of running M on input w?
     - We have no idea
  4. Is $\langle M, w \rangle \in A_{\text{TM}}$?
     - We have no idea
- What do we know about the given TM M and a string w?
  - Technically, we know almost nothing about M and w
  - But we know that if we run TM M on input w, there are three possible outcomes:
    1. M accepts w
    2. M rejects w
    3. M loops indefinitely on w

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_1 =$ "On input x:

  1. Run M on input w
  2. If M accepts w, $accept$. If M rejects w, $reject$."
- What is the language of TM N_1?
  - We cannot answer this question
  - The answer to the above question depends on whether
    - M accepts w,
    - M rejects w, or
    - M loops indefinitely on w

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_1 =$ "On input x:

  1. Run M on input w
  2. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_1 if M accepts w?
  - $L(N_1) = \Sigma^*$ if M accepts w
- Why?

Action	Step 1 of TM N_1	Step 1 Result	Result
Run N_1 on input ε	Run M on input w	*accept*	*accept*
Run N_1 on input 0	Run M on input w	*accept*	*accept*
Run N_1 on input 1	Run M on input w	*accept*	*accept*
Run N_1 on input 00	Run M on input w	*accept*	*accept*
Run N_1 on input 01	Run M on input w	*accept*	*accept*
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- TM N_1 accepts all strings if M accepts w

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_1 =$ "On input x:

  1. Run M on input w
  2. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_1 if M rejects w?
  - $L(N_1) = \emptyset$ if M rejects w
- Why?

Action	Step 1 of TM N_1	Step 1 Result	Result
Run N_1 on input ε	Run M on input w	*reject*	*reject*
Run N_1 on input 0	Run M on input w	*reject*	*reject*
Run N_1 on input 1	Run M on input w	*reject*	*reject*
Run N_1 on input 00	Run M on input w	*reject*	*reject*
Run N_1 on input 01	Run M on input w	*reject*	*reject*
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- TM N_1 rejects all strings if M rejects w

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_1 =$ "On input x:

  1. Run M on input w
  2. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_1 if M loops indefinitely on w?
  - $L(N_1) = \emptyset$ if M loops indefinitely on input w
- Why?

Action	Step 1 of TM N_1	Step 1 Result	Result
Run N_1 on input ε	Run M on input w	M loops indefinitely	N_1 loops indefinitely
Run N_1 on input 0	Run M on input w	M loops indefinitely	N_1 loops indefinitely
Run N_1 on input 1	Run M on input w	M loops indefinitely	N_1 loops indefinitely
Run N_1 on input 00	Run M on input w	M loops indefinitely	N_1 loops indefinitely
Run N_1 on input 01	Run M on input w	M loops indefinitely	N_1 loops indefinitely
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- TM N_1 loops indefinitely on all strings if M loops indefinitely on input w

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_1 =$ "On input x:

  1. Run M on input w
  2. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_1 under the following assumptions:
  - If M accepts $w \rightsquigarrow L(N_1) = \Sigma^*$
  - If M rejects $w \rightsquigarrow L(N_1) = \emptyset$
  - If M loops indefinitely on input $w \rightsquigarrow L(N_1) = \emptyset$

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_2 =$ "On input x:

  1. If $x = \varepsilon$, *accept*
  2. Run M on input w
  3. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_2 under the following assumptions:
  - If M accepts $w \rightsquigarrow L(N_2) = \Sigma^*$
  - If M rejects $w \rightsquigarrow L(N_2) = \{\varepsilon\}$
  - If M loops indefinitely on input $w \rightsquigarrow L(N_2) = \{\varepsilon\}$

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_3 =$ "On input x:

  1. If $x = \varepsilon$, Run M on input w and $accept$ if M does.
  2. $reject$."
- What is the language of TM N_3 under the following assumptions:
  - If M accepts $w \rightsquigarrow L(N_3) = \{\varepsilon\}$
  - If M rejects $w \rightsquigarrow L(N_3) = \emptyset$
  - If M loops indefinitely on input $w \rightsquigarrow L(N_3) = \emptyset$

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_4 =$ "On input x:

  1. If $x = \varepsilon$, Run M on input w. if M accepts w, $accept$. Otherwise, $reject$.
  2. $accept$."
- What is the language of TM N_4 under the following assumptions:
  - If M accepts $w \rightsquigarrow L(N_4) = \Sigma^*$
  - If M rejects $w \rightsquigarrow L(N_4) = \Sigma^* - \{\varepsilon\}$
  - If M loops indefinitely on input $w \rightsquigarrow L(N_4) = \Sigma^* - \{\varepsilon\}$

Turing Machine

- Suppose we have a TM M and a string w
- Consider the following Turing machine:

  $N_5 =$ "On input x:

  1. If x is in the form of 0^n1^n, *accept*
  2. If x is not in the form of 0^n1^n, run M on input w
  3. If M accepts w, *accept*. If M rejects w, *reject*."
- What is the language of TM N_5 under the following assumptions:
  - If M accepts $w \rightsquigarrow L(N_5) = \Sigma^*$
  - If M rejects $w \rightsquigarrow L(N_5) = \{0^n1^n \mid n \geq 0\}$
  - If M loops indefinitely on input $w \rightsquigarrow L(N_5) = \{0^n1^n \mid n \geq 0\}$

Conclusion

- Again, given a TM M and a string w, we know almost nothing about them
- If we construct a TM N that run M on input w, the behavior or language of TM N depends on whether
  1. M accepts w
  2. M rejects w
  3. M loops indefinitely on input w
- What you have seen are some example of turning a TM M and a string w into a new TM
  - Reducibility